Experiment 06

Title

Comparison of Classification Models on MNIST Dataset

SAP ID:	590013857
Dataset:	`https://www.kaggle.com/datasets/hojjatk/mnist-dataset`

AIM

To implement and compare multiple classification models on the MNIST dataset and evaluate their performance using standard metrics.

THEORY

Image classification is a supervised learning task where the model predicts labels for image data.

Key concepts:

- **Classification Models**: Logistic Regression, Decision Tree, Random Forest, SVM, k-NN
- **Evaluation Metrics**: Accuracy, Precision, Recall, F1-score
- **Confusion Matrix**: Measures prediction performance for each class
- **Feature Scaling**: Important for models like SVM and Logistic Regression
- **High-dimensional Data**: MNIST images contain 784 features (pixels)

Commands/Code Used

```
# Core steps

# Load MNIST dataset (from local IDX files)

# Preprocessing
# - Subset selection
# - Feature scaling

# Models used
models = [
    LogisticRegression(),
    DecisionTreeClassifier(),
    RandomForestClassifier(),
    SVC(),
    KNeighborsClassifier()
]
```

```

# Evaluation metrics:
# Accuracy, Precision, Recall, F1-score

# Visualization:
# - Confusion matrices
# - Model comparison bar chart
```

Results

1. Confusion Matrices

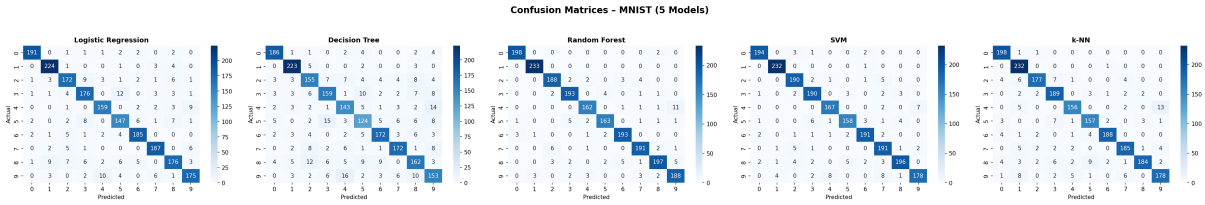

Figure 1: Confusion matrices of 5 classification models

2. Model Performance Comparison

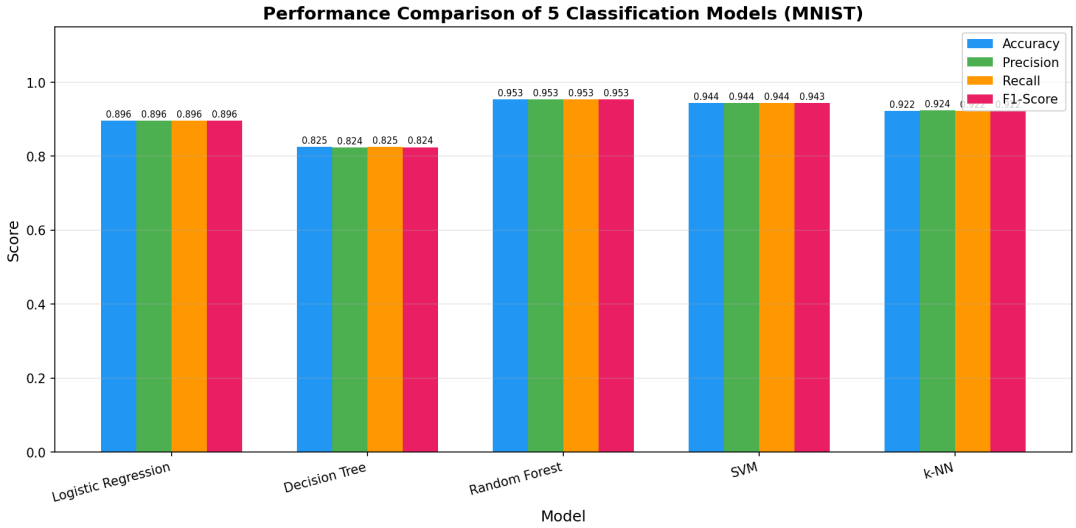

Figure 2: Accuracy, Precision, Recall, and F1-score comparison

Conclusion

In this experiment, five classification models were applied to the MNIST dataset. The results showed that ensemble methods like Random Forest and advanced models like SVM performed better compared to simpler models like Decision Tree.

Confusion matrices provided detailed insights into classification errors, while performance metrics helped compare models effectively.

This experiment demonstrates the importance of model selection and evaluation in classification tasks involving high-dimensional image data.